

Limits



Direct substitution

- 1 Find $\lim_{x \rightarrow 3} (2x^2 - x + 1)$.
 - 2 Find $\lim_{x \rightarrow -1} (x^3 + 4x)$.
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Indeterminate forms: factor and cancel

- 3 Find $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.
 - 4 Find $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2}$.
 - 5 Explain why you cannot find $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$ by direct substitution alone.
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Limits at infinity for rational functions

- 6 Find $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{5x^2 - 1}$.
- 7 Find $\lim_{x \rightarrow \infty} \frac{4x + 1}{2x^2 + 3}$.
- 8 Explain what happens to $\lim_{x \rightarrow \infty} \frac{5x^3 - 2}{x^2 + 1}$, without computing an exact value.